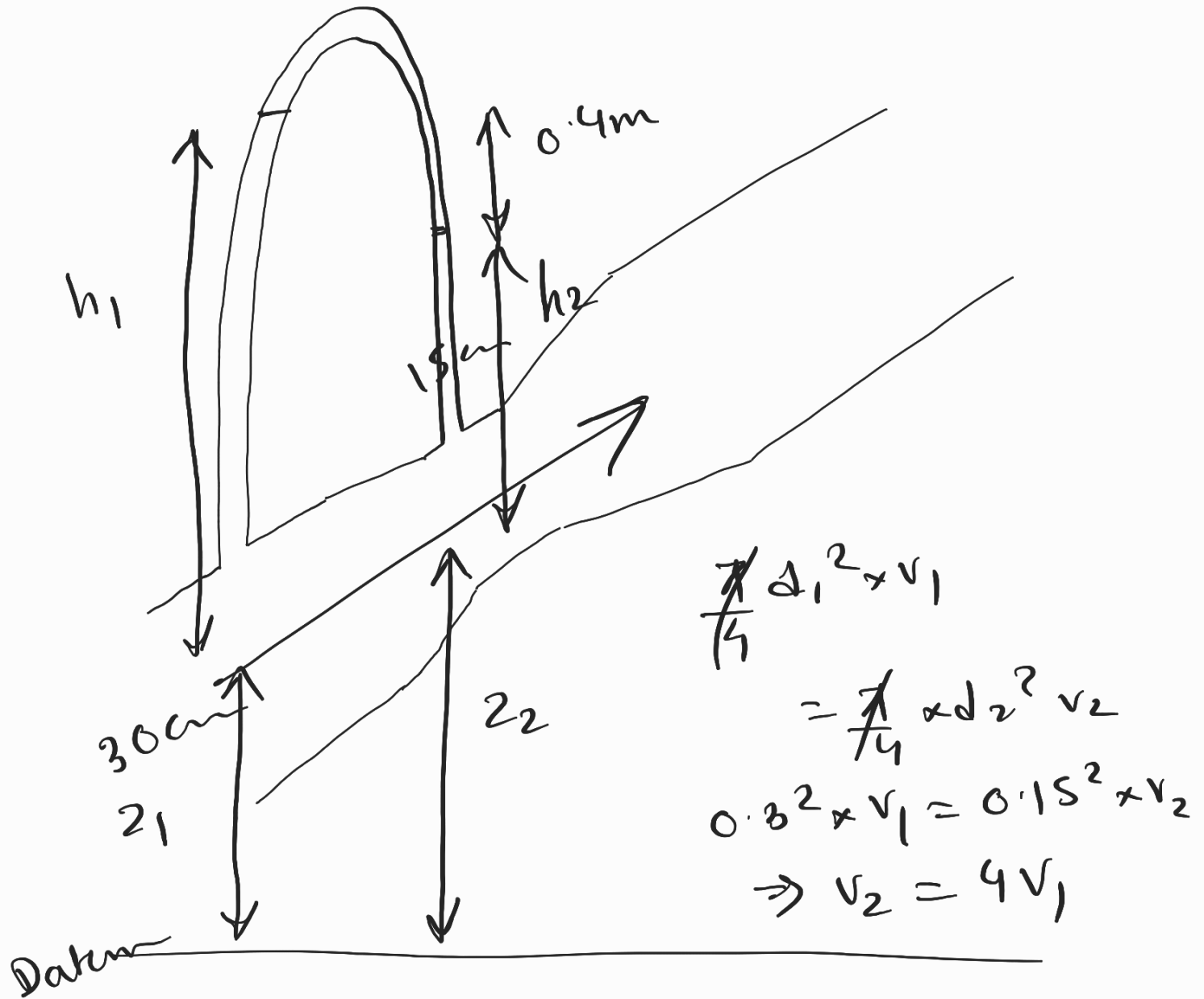




$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + \frac{0.3v_1^2}{2g}$$

$$\frac{v_2^2}{2g} - \frac{0.7v_1^2}{2g} = z_1 - z_2 + \frac{P_1 - P_2}{\rho g}$$

$$\Rightarrow \frac{16v_1^2}{2g} - \frac{0.7v_1^2}{2g} = z_1 - z_2 + \frac{P_1 - P_2}{\rho g}$$

$$\frac{15.3 v_1^2}{2g} = z_1 - z_2 + \frac{p_1 - p_2}{\rho g}$$

$$p_1 - h_1 \rho g = p_2 - h_2 \rho g$$

$$- 0.4 \times 0.8 \times \rho g$$

$$\Rightarrow p_1 - p_2 = h_1 \rho g - h_2 \rho g$$

$$= 0.32 \rho g$$

$$\Rightarrow \boxed{\frac{p_1 - p_2}{\rho g} = h_1 - h_2 - 0.32}$$

$$z_1 + h_1 = z_2 + h_2 + 0.4$$

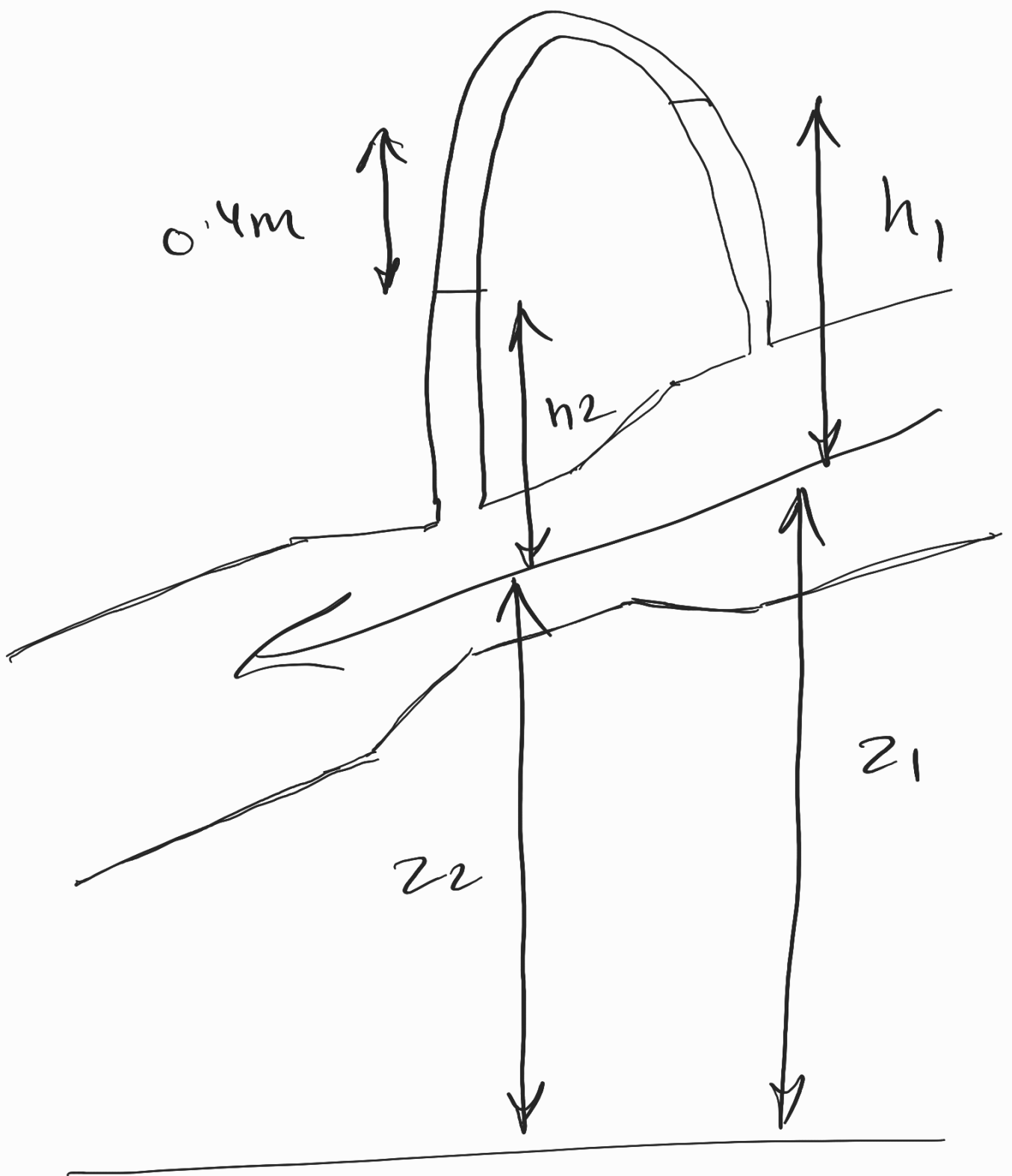
$$\Rightarrow z_1 - z_2 = h_2 - h_1 + 0.4$$

$$\frac{15.3 v_1^2}{2g} = h_2 - h_1 + 0.4$$

$$+ h_1 - h_2 - 0.32$$

$$= 0.08$$

$$\Rightarrow v_1 = 0.32 \text{ m/s} \Rightarrow Q = \frac{\pi}{4} \times 0.3^2 \times 0.32 = 0.0226 \text{ m}^3/\text{s}$$



$$z_1 + \frac{p_1}{\rho g} + \frac{v_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + 0.3 \frac{v_1^2}{2g}$$

$$\Rightarrow \frac{v_2^2}{2g} - 0.7 \frac{v_1^2}{2g} = z_1 - z_2 + \frac{p_1 - p_2}{\rho g}$$

$$p_1 - h_1 \rho g = p_2 - h_2 \rho g - 0.4 \times 0.8 \times \rho g$$

$$\Rightarrow p_1 - p_2 = h_1 \rho g - h_2 \rho g - 0.32 \rho g$$

$$\Rightarrow \frac{p_1 - p_2}{\rho g} = h_1 - h_2 - 0.32$$

$$z_1 + h_1 = z_2 + h_2 + 0.4$$

$$\Rightarrow z_1 - z_2 = h_2 - h_1 + 0.4$$

$$\begin{aligned} \frac{15.3 v_1^2}{2g} &= h_2 - h_1 + 0.4 \\ &+ h_1 - h_2 - 0.32 \\ &= \underline{0.08} \end{aligned}$$